

# Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

## References

- [jqassistant](#)
- [Neo4j Python Driver](#)

## Node Labels

### Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

|    | nodeLabels                                         | nodesWithThatLabels | nodesWithThatLabelsPercent |
|----|----------------------------------------------------|---------------------|----------------------------|
| 0  | [Git, Update, Change]                              | 259019              | 51.786503                  |
| 1  | [Git, Change, Create]                              | 82087               | 16.411918                  |
| 2  | [Git, Change, Delete]                              | 50533               | 10.103226                  |
| 3  | [Git, Change, Rename]                              | 24069               | 4.812193                   |
| 4  | [File, Git]                                        | 16055               | 3.209928                   |
| 5  | [Git, Commit]                                      | 15712               | 3.141351                   |
| 6  | [Java, ByteCode, Parameter]                        | 9258                | 1.850982                   |
| 7  | [Java, ByteCode, Member, Method]                   | 8937                | 1.786803                   |
| 8  | [Java, ByteCode, Bound]                            | 6308                | 1.261179                   |
| 9  | [Java, Value, ByteCode, Annotation]                | 5351                | 1.069843                   |
| 10 | [Java, ByteCode, ParameterizedType, Bound]         | 4854                | 0.970476                   |
| 11 | [Xml, Element]                                     | 2404                | 0.480639                   |
| 12 | [Java, ByteCode, Member, Field]                    | 2213                | 0.442452                   |
| 13 | [Xml, Text]                                        | 1612                | 0.322292                   |
| 14 | [Java, ByteCode, Member, Constructor, Method]      | 1585                | 0.316894                   |
| 15 | [Java, ByteCode, Bound, WildcardType]              | 969                 | 0.193735                   |
| 16 | [Java, ByteCode, Member, Method, Lambda]           | 851                 | 0.170143                   |
| 17 | [Java, ByteCode, Bound, TypeVariable]              | 831                 | 0.166145                   |
| 18 | [Type, File, Java, ByteCode, JavaType]             | 689                 | 0.137754                   |
| 19 | [Type, File, Java, Class, ByteCode]                | 651                 | 0.130157                   |
| 20 | [Json, Key]                                        | 544                 | 0.108764                   |
| 21 | [Value, Json, Scalar]                              | 534                 | 0.106764                   |
| 22 | [Git, Change, Copy]                                | 478                 | 0.095568                   |
| 23 | [Type, File, Java, ByteCode, ResolvedDuplicate...] | 467                 | 0.093369                   |
| 24 | [Java, ByteCode, Member, Method, GenericDeclar...] | 456                 | 0.091170                   |
| 25 | [Java, Value, ByteCode, Primitive]                 | 326                 | 0.065178                   |
| 26 | [Author, Git, Person]                              | 303                 | 0.060580                   |
| 27 | [Type, File, Java, ByteCode, ExternalType]         | 302                 | 0.060380                   |
| 28 | [Committer, Git, Person]                           | 246                 | 0.049184                   |
| 29 | [Maven, Dependency]                                | 194                 | 0.038787                   |

## Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

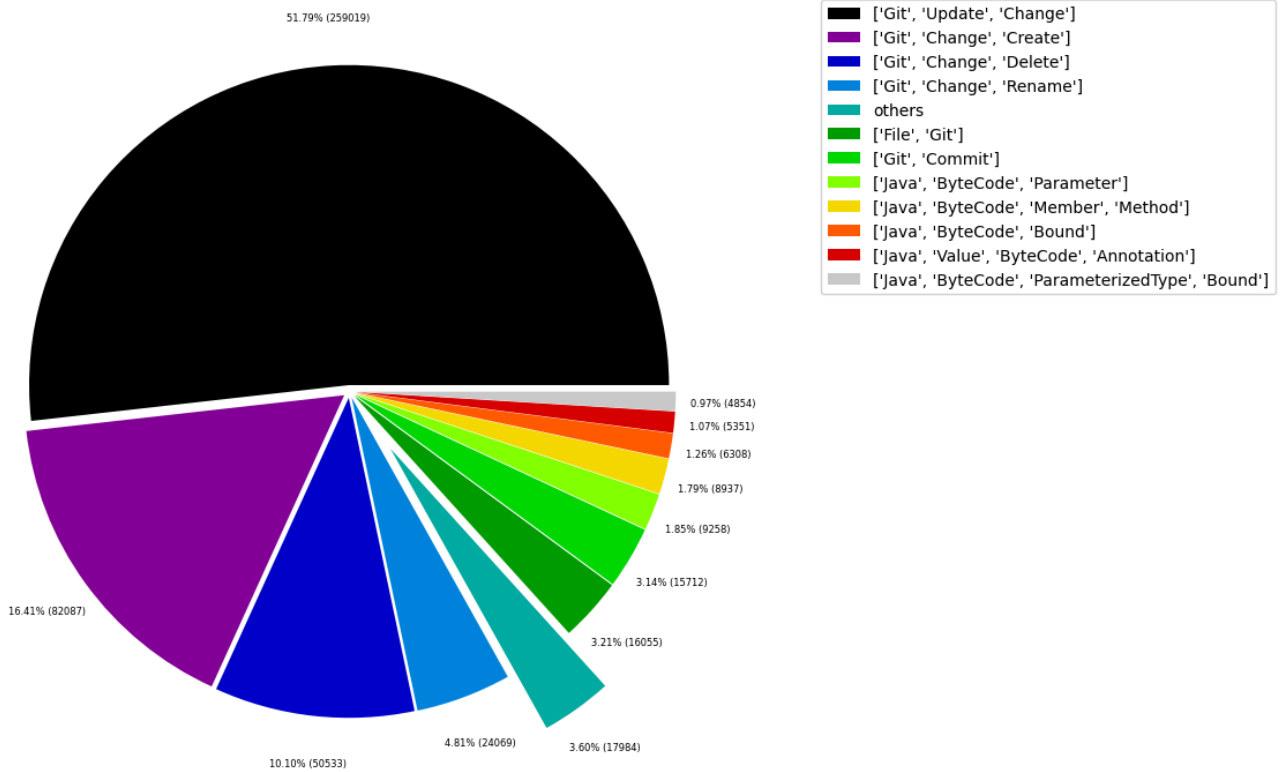


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

|    | nodeLabels                                        | nodesWithThatLabels | nodesWithThatLabelsPercent |
|----|---------------------------------------------------|---------------------|----------------------------|
| 0  | [Analyze, Task, jQAssistant]                      | 1                   | 0.000200                   |
| 1  | [Git, Branch]                                     | 1                   | 0.000200                   |
| 2  | [Repository, File, Git]                           | 1                   | 0.000200                   |
| 3  | [File, Json]                                      | 2                   | 0.000400                   |
| 4  | [File]                                            | 3                   | 0.000600                   |
| 5  | [Maven, Exclusion]                                | 4                   | 0.000800                   |
| 6  | [Type, File, Java, ByteCode, GenericDeclaratio... | 4                   | 0.000800                   |
| 7  | [Value, Array, Json]                              | 8                   | 0.001599                   |
| 8  | [Type, File, Java, ByteCode, Throwable, Extern... | 8                   | 0.001599                   |
| 9  | [Java, ByteCode, Member, Constructor, Method, ... | 9                   | 0.001799                   |
| 10 | [Type, File, Java, ByteCode, Throwable, Resolv... | 11                  | 0.002199                   |
| 11 | [Maven, Scm]                                      | 11                  | 0.002199                   |
| 12 | [File, Maven, Xml, Pom, Document]                 | 11                  | 0.002199                   |
| 13 | [Type, File, Java, ByteCode, Void]                | 11                  | 0.002199                   |
| 14 | [Java, ManifestSection]                           | 11                  | 0.002199                   |
| 15 | [File, Java, Manifest]                            | 11                  | 0.002199                   |
| 16 | [File, Artifact, Jar, Archive, Zip, Java]         | 11                  | 0.002199                   |
| 17 | [File, Java, ServiceLoader]                       | 12                  | 0.002399                   |
| 18 | [File, Java, Properties]                          | 13                  | 0.002599                   |
| 19 | [Maven, ExecutionGoal]                            | 13                  | 0.002599                   |
| 20 | [Maven, PluginExecution]                          | 13                  | 0.002599                   |
| 21 | [Type, File, Java, ByteCode, Enum]                | 17                  | 0.003399                   |
| 22 | [jQAssistant, Rule, Concept]                      | 19                  | 0.003799                   |
| 23 | [Maven, Configuration]                            | 20                  | 0.003999                   |
| 24 | [Maven, Plugin]                                   | 20                  | 0.003999                   |
| 25 | [Xml, Attribute]                                  | 22                  | 0.004399                   |
| 26 | [Type, File, Java, ByteCode, PrimitiveType]       | 39                  | 0.007797                   |
| 27 | [Type, File, Java, ByteCode, Annotation]          | 42                  | 0.008397                   |
| 28 | [Type, File, Java, Class, ByteCode, Throwable]    | 42                  | 0.008397                   |
| 29 | [Xml, Namespace]                                  | 44                  | 0.008797                   |

## Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (less than 0.5% overall)

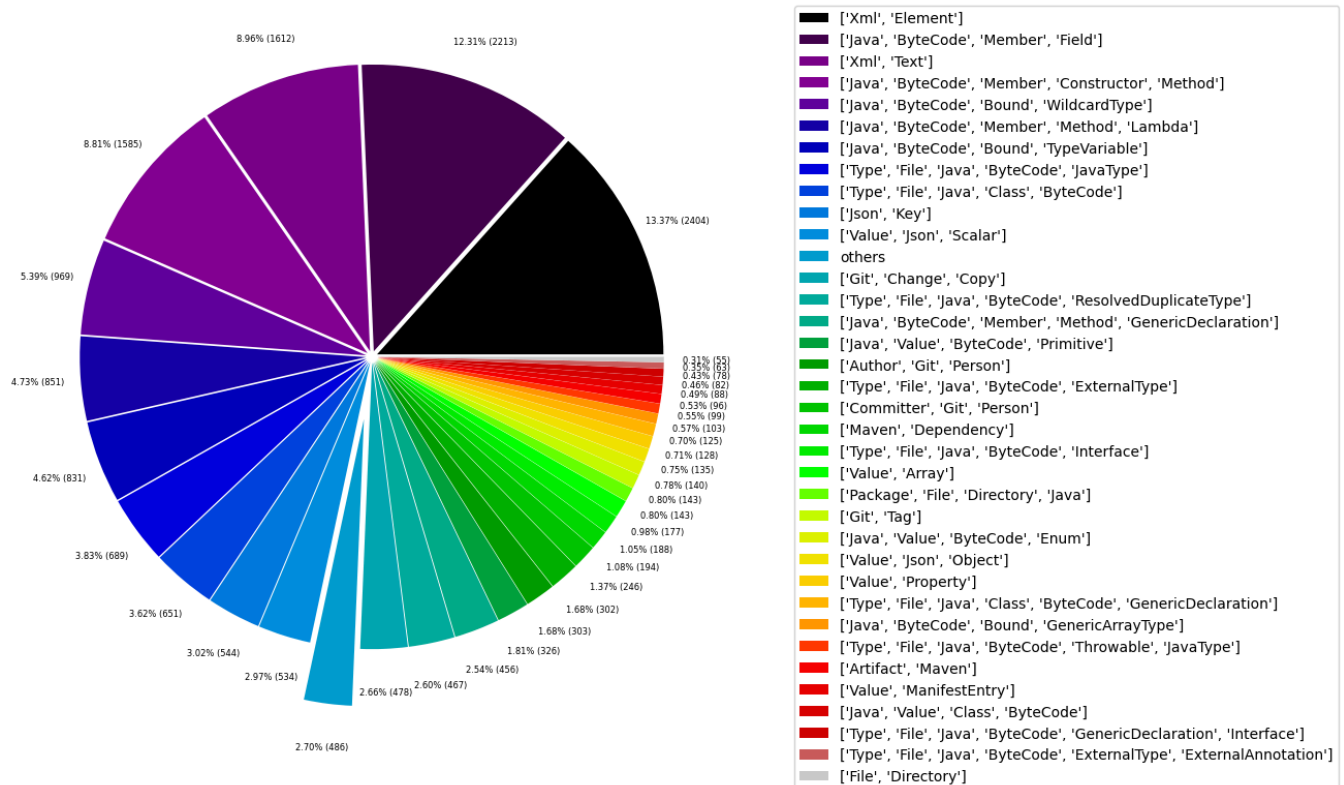


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

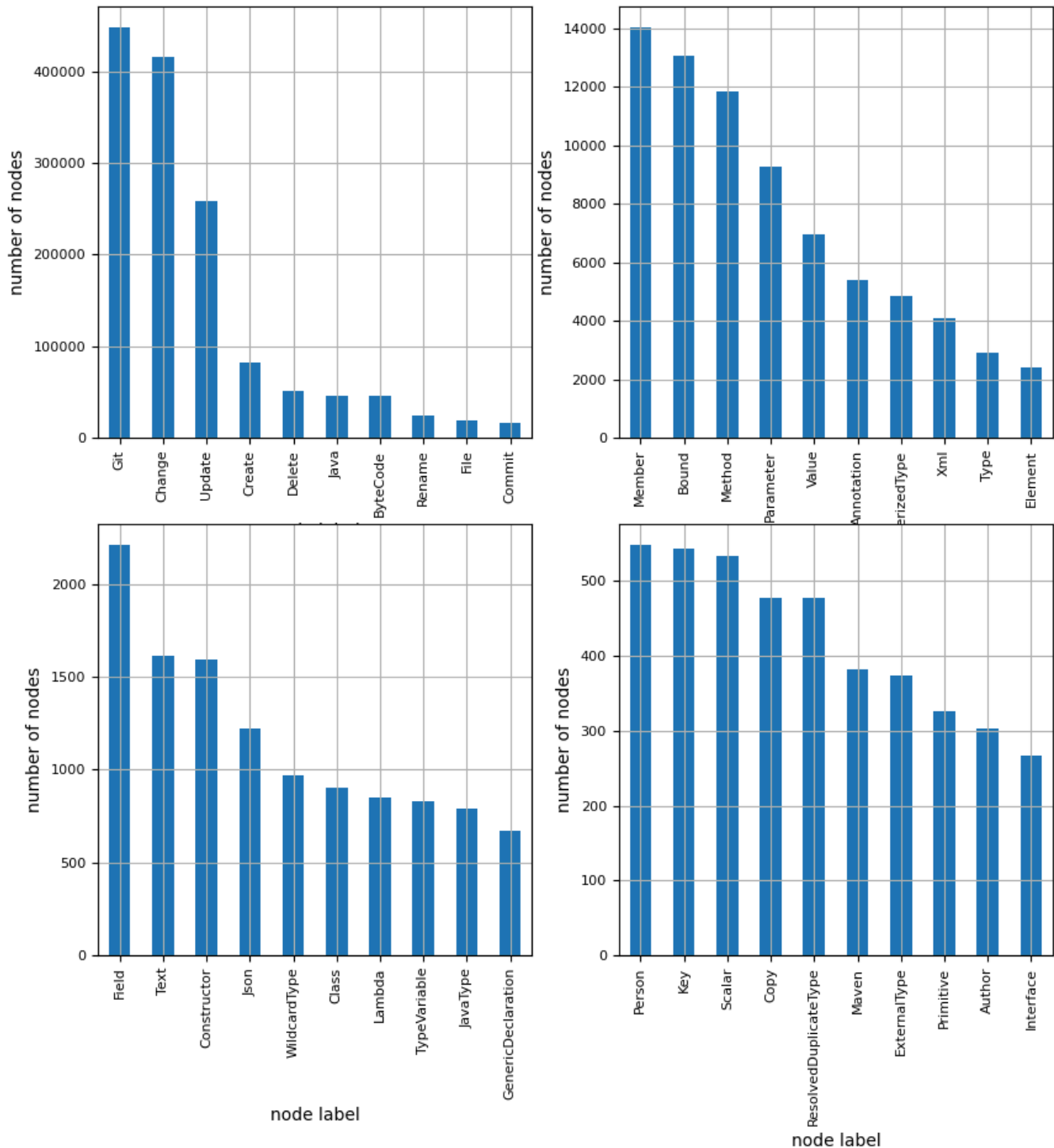
|    | nodeLabel             | nodesWithThatLabel | nodesWithThatLabelPercent |
|----|-----------------------|--------------------|---------------------------|
| 0  | Git                   | 448647             | 89.699440                 |
| 1  | Change                | 416186             | 83.209408                 |
| 2  | Update                | 259019             | 51.786503                 |
| 3  | Create                | 82087              | 16.411918                 |
| 4  | Delete                | 50533              | 10.103226                 |
| 5  | Java                  | 45362              | 9.069371                  |
| 6  | ByteCode              | 45161              | 9.029184                  |
| 7  | Rename                | 24069              | 4.812193                  |
| 8  | File                  | 19205              | 3.839718                  |
| 9  | Commit                | 15712              | 3.141351                  |
| 10 | Member                | 14051              | 2.809262                  |
| 11 | Bound                 | 13065              | 2.612128                  |
| 12 | Method                | 11838              | 2.366809                  |
| 13 | Parameter             | 9258               | 1.850982                  |
| 14 | Value                 | 6969               | 1.393335                  |
| 15 | Annotation            | 5393               | 1.078240                  |
| 16 | ParameterizedType     | 4854               | 0.970476                  |
| 17 | Xml                   | 4093               | 0.818327                  |
| 18 | Type                  | 2888               | 0.577407                  |
| 19 | Element               | 2404               | 0.480639                  |
| 20 | Field                 | 2213               | 0.442452                  |
| 21 | Text                  | 1612               | 0.322292                  |
| 22 | Constructor           | 1594               | 0.318694                  |
| 23 | Json                  | 1223               | 0.244518                  |
| 24 | WildcardType          | 969                | 0.193735                  |
| 25 | Class                 | 900                | 0.179940                  |
| 26 | Lambda                | 851                | 0.170143                  |
| 27 | TypeVariable          | 831                | 0.166145                  |
| 28 | JavaType              | 788                | 0.157547                  |
| 29 | GenericDeclaration    | 672                | 0.134355                  |
| 30 | Person                | 549                | 0.109763                  |
| 31 | Key                   | 544                | 0.108764                  |
| 32 | Scalar                | 534                | 0.106764                  |
| 33 | Copy                  | 478                | 0.095568                  |
| 34 | ResolvedDuplicateType | 478                | 0.095568                  |
| 35 | Maven                 | 382                | 0.076374                  |
| 36 | ExternalType          | 373                | 0.074575                  |
| 37 | Primitive             | 326                | 0.065178                  |
| 38 | Author                | 303                | 0.060580                  |
| 39 | Interface             | 266                | 0.053182                  |

## Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

<Figure size 640x480 with 0 Axes>

## Node count by label



## Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1553446

|    | relationshipType         | nodesWithThatRelationshipType | nodesWithThatRelationshipTypePercent |
|----|--------------------------|-------------------------------|--------------------------------------|
| 0  | CONTAINS_CHANGE          | 416186                        | 26.791147                            |
| 1  | MODIFIES                 | 416186                        | 26.791147                            |
| 2  | UPDATES                  | 259019                        | 16.673834                            |
| 3  | CREATES                  | 106634                        | 6.864352                             |
| 4  | DELETES                  | 74602                         | 4.802356                             |
| 5  | COMMITTED                | 31424                         | 2.022858                             |
| 6  | RENAMES                  | 24069                         | 1.549394                             |
| 7  | INVOKES                  | 22322                         | 1.436934                             |
| 8  | HAS_PARENT               | 19094                         | 1.229138                             |
| 9  | OF_TYPE                  | 18557                         | 1.194570                             |
| 10 | DEPENDS_ON               | 17631                         | 1.134961                             |
| 11 | HAS_FILE                 | 16055                         | 1.033509                             |
| 12 | HAS_COMMIT               | 15712                         | 1.011429                             |
| 13 | DECLARES                 | 14393                         | 0.926521                             |
| 14 | OF_RAW_TYPE              | 13136                         | 0.845604                             |
| 15 | HAS                      | 9882                          | 0.636134                             |
| 16 | HAS_NEW_NAME             | 9396                          | 0.604849                             |
| 17 | RETURNS                  | 8126                          | 0.523095                             |
| 18 | READS                    | 5948                          | 0.382891                             |
| 19 | HAS_ACTUAL_TYPE_ARGUMENT | 5758                          | 0.370660                             |
| 20 | ANNOTATED_BY             | 5350                          | 0.344396                             |
| 21 | OF_GENERIC_TYPE          | 4397                          | 0.283048                             |
| 22 | CHANGED_TOGETHER_WITH    | 3982                          | 0.256333                             |
| 23 | SIMILAR                  | 3380                          | 0.217581                             |
| 24 | CONTAINS                 | 3194                          | 0.205607                             |
| 25 | RESOLVES_TO              | 2611                          | 0.168078                             |
| 26 | RETURNS_GENERIC          | 2590                          | 0.166726                             |
| 27 | HAS_FIRST_CHILD          | 2404                          | 0.154753                             |
| 28 | HAS_LAST_CHILD           | 2404                          | 0.154753                             |
| 29 | HAS_ELEMENT              | 2382                          | 0.153337                             |

### Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

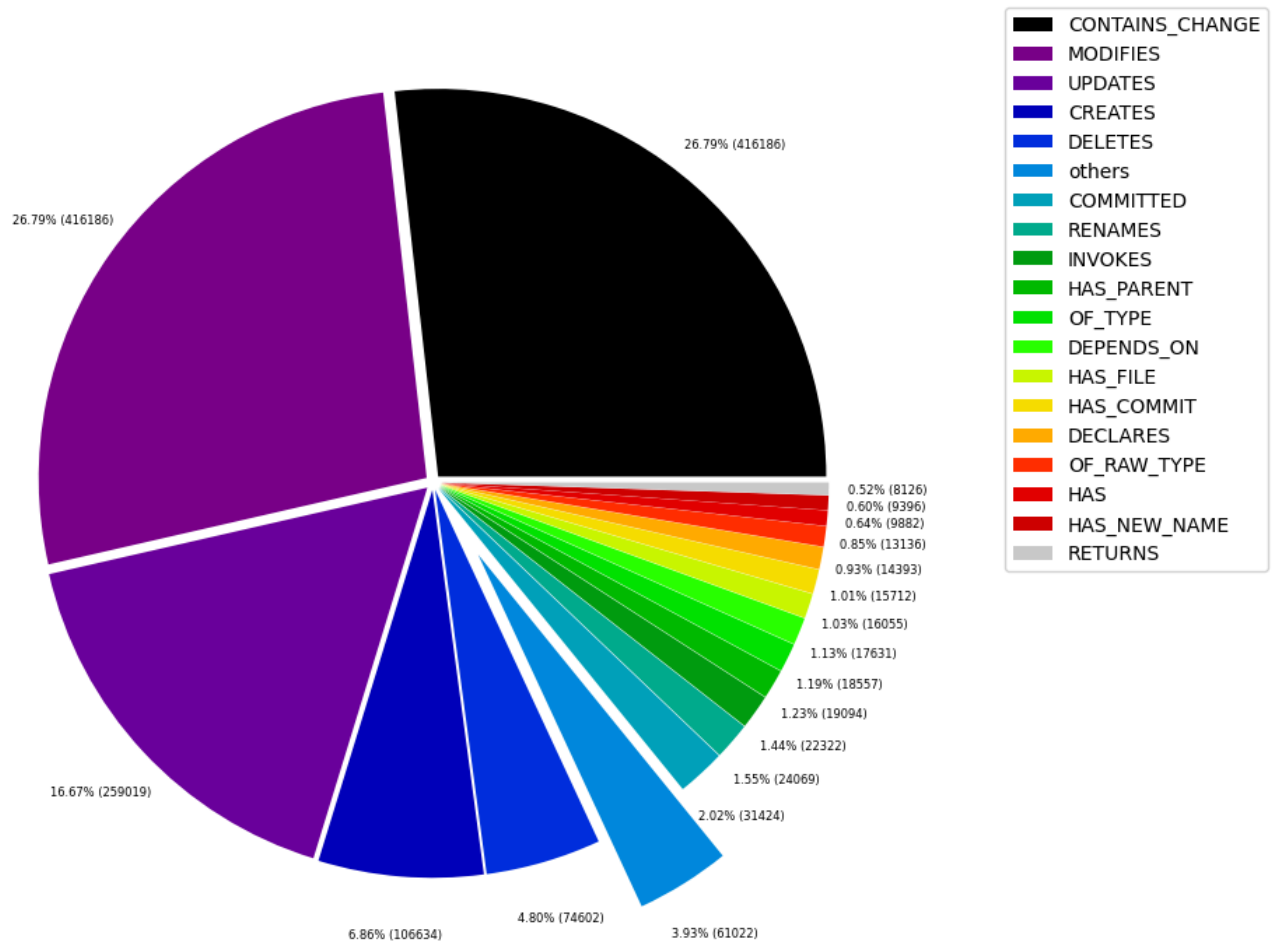


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

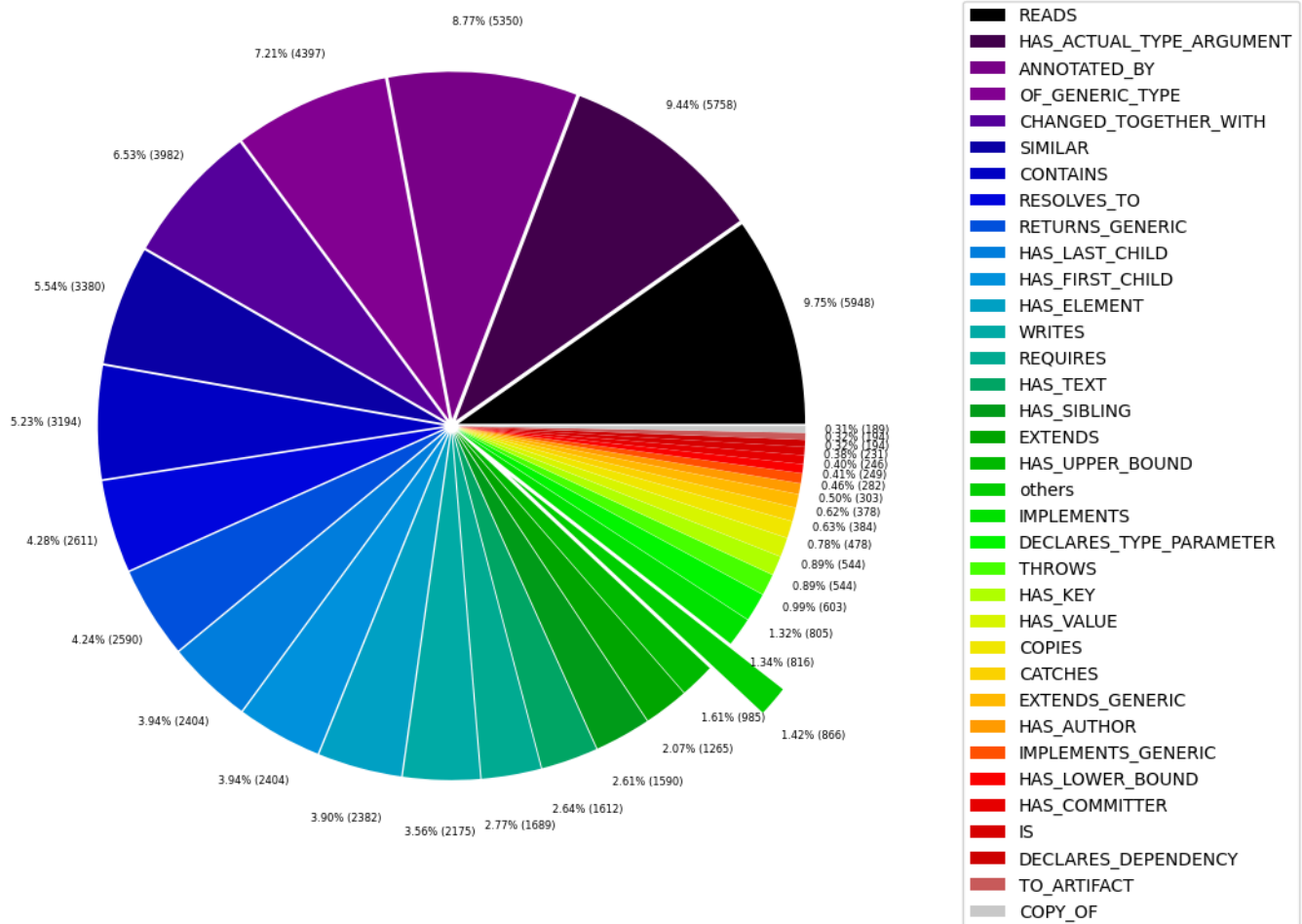
|    | relationshipType        | nodesWithThatRelationshipType | nodesWithThatRelationshipTypePercent |
|----|-------------------------|-------------------------------|--------------------------------------|
| 0  | HAS_PROPERTY            | 1                             | 0.000064                             |
| 1  | HAS_BRANCH              | 1                             | 0.000064                             |
| 2  | HAS_HEAD                | 2                             | 0.000129                             |
| 3  | THROWS_GENERIC          | 3                             | 0.000193                             |
| 4  | EXCLUDES                | 4                             | 0.000257                             |
| 5  | HAS_SCM                 | 11                            | 0.000708                             |
| 6  | DESCRIBES               | 11                            | 0.000708                             |
| 7  | HAS_ROOT_ELEMENT        | 13                            | 0.000837                             |
| 8  | HAS_GOAL                | 13                            | 0.000837                             |
| 9  | HAS_EXECUTION           | 13                            | 0.000837                             |
| 10 | INCLUDES_CONCEPT        | 19                            | 0.001223                             |
| 11 | IS_ARTIFACT             | 20                            | 0.001287                             |
| 12 | HAS_CONFIGURATION       | 20                            | 0.001287                             |
| 13 | USES_PLUGIN             | 20                            | 0.001287                             |
| 14 | OF_NAMESPACE            | 22                            | 0.001416                             |
| 15 | HAS_ATTRIBUTE           | 22                            | 0.001416                             |
| 16 | REQUIRES_TYPE_PARAMETER | 26                            | 0.001674                             |
| 17 | REQUIRES_CONCEPT        | 28                            | 0.001802                             |
| 18 | DECLARES_NAMESPACE      | 44                            | 0.002832                             |
| 19 | HAS_DEFAULT             | 53                            | 0.003412                             |
| 20 | HAS_COMPONENT_TYPE      | 103                           | 0.006630                             |
| 21 | CONTAINS_VALUE          | 131                           | 0.008433                             |
| 22 | HAS_TAG                 | 143                           | 0.009205                             |
| 23 | ON_COMMIT               | 143                           | 0.009205                             |
| 24 | COPY_OF                 | 189                           | 0.012166                             |
| 25 | TO_ARTIFACT             | 194                           | 0.012488                             |
| 26 | DECLARES_DEPENDENCY     | 194                           | 0.012488                             |
| 27 | IS                      | 231                           | 0.014870                             |
| 28 | HAS_COMMITTER           | 246                           | 0.015836                             |
| 29 | HAS_LOWER_BOUND         | 249                           | 0.016029                             |

## Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



## Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

|    | sourceLabels                               | relationType          | targetLabels                                      | numberOfRelationships | numberOfNodesWithSameLabelsAsSource | numberOfNodes |
|----|--------------------------------------------|-----------------------|---------------------------------------------------|-----------------------|-------------------------------------|---------------|
| 0  | [Git, Commit]                              | CONTAINS_CHANGE       | [Git, Update, Change]                             | 259019                | 15712                               |               |
| 1  | [Git, Update, Change]                      | UPDATES               | [File, Git]                                       | 259019                | 259019                              |               |
| 2  | [Git, Update, Change]                      | MODIFIES              | [File, Git]                                       | 259019                | 259019                              |               |
| 3  | [Git, Commit]                              | CONTAINS_CHANGE       | [Git, Change, Create]                             | 82087                 | 15712                               |               |
| 4  | [Git, Change, Create]                      | MODIFIES              | [File, Git]                                       | 82087                 | 82087                               |               |
| 5  | [Git, Change, Create]                      | CREATES               | [File, Git]                                       | 82087                 | 82087                               |               |
| 6  | [Git, Commit]                              | CONTAINS_CHANGE       | [Git, Change, Delete]                             | 50533                 | 15712                               |               |
| 7  | [Git, Change, Delete]                      | DELETES               | [File, Git]                                       | 50533                 | 50533                               |               |
| 8  | [Git, Change, Delete]                      | MODIFIES              | [File, Git]                                       | 50533                 | 50533                               |               |
| 9  | [Git, Commit]                              | CONTAINS_CHANGE       | [Git, Change, Rename]                             | 24069                 | 15712                               |               |
| 10 | [Git, Change, Rename]                      | CREATES               | [File, Git]                                       | 24069                 | 24069                               |               |
| 11 | [Git, Change, Rename]                      | MODIFIES              | [File, Git]                                       | 24069                 | 24069                               |               |
| 12 | [Git, Change, Rename]                      | RENAMES               | [File, Git]                                       | 24069                 | 24069                               |               |
| 13 | [Git, Change, Rename]                      | DELETES               | [File, Git]                                       | 24069                 | 24069                               |               |
| 14 | [Git, Commit]                              | HAS_PARENT            | [Git, Commit]                                     | 19083                 | 15712                               |               |
| 15 | [Repository, File, Git]                    | HAS_FILE              | [File, Git]                                       | 16055                 | 1                                   |               |
| 16 | [Repository, File, Git]                    | HAS_COMMIT            | [Git, Commit]                                     | 15712                 | 1                                   |               |
| 17 | [Author, Git, Person]                      | COMMITTED             | [Git, Commit]                                     | 15712                 | 303                                 |               |
| 18 | [Committer, Git, Person]                   | COMMITTED             | [Git, Commit]                                     | 15712                 | 246                                 |               |
| 19 | [Java, ByteCode, Member, Method]           | INVOKES               | [Java, ByteCode, Member, Method]                  | 13439                 | 8911                                |               |
| 20 | [File, Git]                                | HAS_NEW_NAME          | [File, Git]                                       | 9396                  | 16055                               |               |
| 21 | [Java, ByteCode, Member, Method]           | HAS                   | [Java, ByteCode, Parameter]                       | 5352                  | 8911                                |               |
| 22 | [Java, ByteCode, Member, Method]           | READS                 | [Java, ByteCode, Member, Field]                   | 5350                  | 8911                                |               |
| 23 | [Java, Value, ByteCode, Annotation]        | OF_TYPE               | [Type, File, Java, ByteCode, ExternalType, Ext... | 5006                  | 5351                                |               |
| 24 | [Java, ByteCode, Parameter]                | OF_TYPE               | [Type, File, Java, ByteCode, JavaType]            | 4114                  | 9258                                |               |
| 25 | [Java, ByteCode, Parameter]                | ANNOTATED_BY          | [Java, Value, ByteCode, Annotation]               | 3805                  | 9258                                |               |
| 26 | [File, Git]                                | CHANGED_TOGETHER_WITH | [File, Git]                                       | 3349                  | 16055                               |               |
| 27 | [Java, ByteCode, ParameterizedType, Bound] | OF_RAW_TYPE           | [Type, File, Java, ByteCode, JavaType]            | 2804                  | 4854                                |               |
| 28 | [Xml, Element]                             | HAS_ELEMENT           | [Xml, Element]                                    | 2382                  | 2404                                |               |
| 29 | [Java, ByteCode, Bound]                    | OF_RAW_TYPE           | [Type, File, Java, ByteCode, JavaType]            | 2260                  | 6308                                |               |

# Graph Density

total\_number\_of\_nodes (vertices): 500167

total\_number\_of\_relationships (edges): 1553446

-> total directed graph density: 6.209647686065667e-06

-> total directed graph density in percent: 0.0006209647686065667